

Title

Metformin Attenuates Aerobic Fitness Gains While Enhancing Fat Oxidation Capacity During Exercise Training in Metabolic Syndrome

Abstract

Twenty-four metformin-naïve adults with metabolic syndrome completed a five-week supervised aerobic training program while receiving either metformin (1000 mg/day, $n = 11$) or placebo ($n = 11$). Peak oxygen uptake improvements showed a modest group difference: the metformin group gained 0.09 L/min compared to 0.28 L/min in the placebo group ($p = 0.048$). Relative to body weight, this difference approached but did not reach statistical significance ($p = 0.095$). In contrast, maximal fat oxidation capacity demonstrated substantially greater augmentation in the metformin group (+0.15 versus +0.03 g/min, $p = 0.001$). Submaximal exercise substrate oxidation patterns similarly favored the metformin group, with significantly elevated fat utilization during moderate-intensity efforts. Perceived exertion ratings were higher in the metformin group during high-intensity intervals ($p = 0.019$). Age-stratified analysis revealed that metformin-induced cardiorespiratory attenuation was more pronounced in participants older than 55 years (correlation $r = -0.86$, $p < 0.001$). These findings suggest that metformin produces a functionally distinct metabolic adaptation during aerobic training, trading aerobic capacity for enhanced lipid utilization and greater perceived effort.